



# Automated line identification for atomic spectroscopy (ALIAS): Application to LIBS imaging data processing

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## ARTICLE INFO

### Keywords:

LIBS elemental imaging  
Automatic line identification  
Data processing

## ABSTRACT

Micro LIBS (laser-induced breakdown spectroscopy) imaging has experienced rapid growth in recent years, enabling the generation of increasingly large and complex datasets. One of the main challenges today lies in data processing, which can be extremely complex and time-consuming. In this article, we propose a new method for automatically identifying emission lines and associated elements in LIBS spectra. This work is part of LIBS imaging data processing, aimed at interpreting averaged spectra from various significant regions of the studied sample. The proposed methodology, named ALIAS (Automated Line Identification for Atomic Spectroscopy), operates by using multiple defined coefficient to find similarities between an acquired spectrum and a theoretical one derived from a simplified plasma model. The comprehensive methodology and the sequential stages of ALIAS—including peak detection, synthetic spectrum generation, intensity threshold determination, similarity coefficient computation, decision-making, and probabilistic assessment—are meticulously detailed. ALIAS's performance is then assessed across several real-world cases that present commonly encountered challenges. This study highlights the efficiency and robustness of the ALIAS method, facilitating a high degree of automation in data processing—an enduring bottleneck that continues to hinder the widespread adoption of LIBS-based imaging technology.

## 1. Introduction

Micro laser-induced breakdown spectroscopy ( $\mu$ LIBS) imaging has experienced remarkable growth in recent years [1–4]. As a now well-established analytical technique,  $\mu$ LIBS imaging is gaining prominence in critical fields such as biomedicine [5–7], the exploration of strategic elements in geoscience [8–10], and the development of advanced materials for future energy applications [11–13], to name but a few. Current instruments are more powerful than ever, capable of rapidly generating vast amounts of spectral data [14,15]. One of the critical aspects of this technology is undoubtedly the data processing stage, which is essential for extracting meaningful insights from these large datasets. However, this step is highly complex, especially given the large volume of spectra, and is currently accessible only to experts in the field. This limitation significantly hinders the broader adoption of the technology by general users [4].

A promising approach currently being explored is the use of statistical analysis methods from chemometrics and machine learning for multivariate data processing. These approaches allow, for example, reducing the dimensionality of a dataset by identifying correlations between spectral variables or the spectra themselves, making the information more accessible and usable. Among these methodologies, principal component analysis [16,17], clustering [18,19], neural networks [11,20], and signal unmixing [21] stand as foundational techniques. More recently, the Interesting Features Finder (IFF) has emerged as a powerful approach, enabling the identification of rare spectral structures within extensive datasets [22]. Although these approaches hold significant potential for the future, they still require adaptation on a case-by-case basis, depending on specific needs and the structure of the dataset [4].

In general, we employ a conventional univariate processing method to generate elemental images through simple integration of the lines of

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<https://doi.org/10.1016/j.sab.2025.107255>

Received 7 February 2025; Received in revised form 10 June 2025; Accepted 11 June 2025

Available online 16 June 2025

0584-8547/© 2025 Published by Elsevier B.V.

interest [23]. This is, moreover, a strategy that is not unique to LIBS, as it has always been employed from the early stages by all spectroscopic imaging techniques. The key steps involve detecting all present elements and selecting corresponding specific emission lines, free from spectral interferences. To achieve this, we represent the dataset with its averaged spectrum, initially computed across and subsequently refined through iterative calculations over the distinct mineral phases of the sample [17,24]. This iterative procedure allows a step-by-step approach: first, gaining a global overview of the analysis (i.e., major elements), and then, by defining masks and associated averaged spectra, examining minor and trace elements within each mineral phase of the sample. At each stage (i.e., for each averaged spectrum), it is necessary to manually label all lines and then identify corresponding elements present in the spectrum. This step can be labor-intensive and, more importantly, represents a significant potential source of error. Our aim in this article is to address this issue by proposing an automated methodology for line identification.

The automatic identification of spectral lines in LIBS spectra has long been a major challenge. LIBS spectra are known for their high complexity, influenced by various factors such as matrix effects, wide measurement dynamics, noise and baseline variability, self-absorption, spectral interferences, and the dense spectral features of certain elements such as transition metals (Fe, Ti, Mo, Ni, etc.). Significant progress has recently been achieved in the automated identification of spectral lines in LIBS analysis. Mateo et al. [25] proposed a line identification method based on wavelength matching, which involves comparing the wavelengths of emission lines with those listed in the NIST database [26]. Amato et al. introduced a text retrieval algorithm grounded in the vector space model, treating an element as a 'document', a peak as a 'word', and the sample as a 'query' [27]. The matching relevance was determined by calculating the cosine similarity between the query vector and the document vector, followed by assigning an appropriate weight value. A novel identification approach was suggested by Labutin et al. [28], who utilized the correlation between experimental and synthetic model spectra. Their algorithm comprises three steps: simulating a set of spectra corresponding to different temperatures ( $T$ ) and electron number densities ( $n_e$ ), searching for the best-correlated pair between a model spectrum and an experimental spectrum, and finally attributing lines to the identified peaks. Rosas-Roman et al. introduced a similitude coefficient, called the wavelength similarity coefficient, to preserve the relative relevance of line emission intensities [29]. Furthermore, the authors demonstrated that for identification purposes, a simplified plasma model was sufficient to accurately predict the relative emission intensities in LIBS plasmas. Zhang et al. proposed a simple method for element recognition that involves comparing vectors created from the quantities of spectral peaks. This approach enabled accurate identification of the principal elements, regardless of any wavelength shifts [30]. More recently, Gajarska et al. proposed an algorithm that uses spectral fingerprints of elements configured as comb-like filters, which are then correlated with experimental spectra. Such algorithm has the great advantage of maintaining accuracy even with deviations from ideal calibration caused by instrumental fluctuations [31].

All these methods offer innovative solutions and generally deliver good performance. We have tested most of them in conjunction with relative line intensities provided by atomic databases. These methods include correlation analysis, wavelength matching, and spectral similarity metrics. However, none of them, when used alone, proved robust enough for the reliable processing of our spectra. In all cases, we encountered several significant issues, such as well-known emission lines remaining unidentified, lines incorrectly attributed to implausible elements (particularly those with a high density of spectral lines), and, most notably, a strong sensitivity to spectral interferences. These limitations stem from specific characteristics of the LIBS imaging systems used in our study. In  $\mu$ LIBS imaging, to ensure a balance between versatility, sensitivity, spectral resolution, and acquisition speed, we

rely on Czerny-Turner spectrometers coupled with ICCD cameras [32]. This configuration restricts us to narrow spectral windows, typically ranging from 20 to 80 nm, depending on the diffraction grating. Moreover, processing averaged spectra requires handling a wide dynamic range—from lines barely distinguishable from the background to highly intense peaks. The complexity of certain samples—particularly geological ones—further complicates the task, as a large number of elements may be simultaneously present in a single averaged spectrum, generating numerous potential interferences. Working within these narrow spectral ranges and under such high dynamic range conditions introduces additional challenges. For example, the most intense lines of some elements (often the most reliable in databases) may fall outside the acquisition window. In some cases, only a single emission line is observed or identifiable for a given element, making conventional identification metrics ineffective. A typical example arises when a weak line from element A overlaps with a strong line from element B. In such cases, correlation or similarity coefficients may return misleadingly low values, preventing the correct detection of element A.

In this article, we introduce a methodology called ALIAS (Automated Line Identification for Atomic Spectroscopy), designed to label spectral lines and identify the corresponding elements. This method combines several approaches, including some adapted from the literature, to enhance robustness and enable adaptability across typical situations encountered in LIBS imaging—such as wide dynamic range, high spectral complexity, and a large number of detected elements. As previously discussed, our goal is not to identify lines in each single-shot spectrum of an imaging dataset but to produce a comprehensive identification of high signal to noise ratio (SNR) averaged spectra generated during the data processing stage. In addition to the limited spectral range, these averaged spectra often present a high dynamic range. We will evaluate the ALIAS method's on two well-characterized sample types: a rock with a high iron concentration and a NIST glass (SRM1412) containing numerous elements. Finally, a detailed application of ALIAS to imaging data processing will be presented, using a cassiterite rock sample as a case study.

## 2. Material and methods

### 2.1. Instrumentation

The general protocol and the LIBS instrument used for this study have already been described in previous works [24,32]. In summary, a Nd:YAG laser (Centurion, Quantel Laser) was used at its fundamental wavelength, operating at a repetition rate of 100 Hz and a pulse energy of 1 mJ, controlled by an attenuator. The laser pulses were focused on the sample surface using a  $15\times$  magnification objective (LMM-15 $\times$ -P01, Thorlabs). Samples were systematically moved on a motorized XYZ stage to enable pixel-by-pixel scanning, with an adjustable intershot distance ranging from 10 to 100  $\mu$ m. Both the laser pulse energy and the focal distance relative to the sample surface were closely controlled throughout the experiments. LIBS measurements were conducted under ambient conditions, including room temperature and atmospheric pressure, with an argon flow of 1.0 L/min directed at the plasma region for confinement and stability. The emitted plasma radiation was collected through a lens-fiber system connected to a Czerny-Turner spectrometer (Shamrock 500, Andor Technology) equipped with an intensified charge-coupled device (ICCD) camera (Istar 340 T, Andor Technology) synchronized with the laser's Q-switch. This rectangular camera (2048  $\times$  512 pixels) is equipped with a 25 mm intensifier, providing access to approximately 1900 wavelength channels. The entrance slit of the spectrometer was set to 35  $\mu$ m. ICCD acquisitions were performed without any gain and with a delay and gate of 1  $\mu$ s and 2  $\mu$ s, respectively. Measurements for this study were conducted using either a 600 lines/mm grating, offering an 80 nm spectral range with a resolution of 0.015 nm, or a 2400 lines/mm grating, which enables a 20 nm spectral range with a higher resolution of 0.04 nm. The spectra

processed afterward in this study represent averages of a large number of single-shot spectra (typically over 20,000).

## 2.2. Samples

Two samples were selected to evaluate the ALIAS performance. The first sample is a geologically well-characterized rock composed of multiple two Cu-Fe-rich sulfide mineral phases, including chalcopyrite ( $\text{CuFeS}_2$ ) and tennantite (ideal formula of  $\text{Cu}_6[\text{Cu}_4(\text{Fe,Zn})_2]\text{As}_4\text{S}_{13}$ ). This sample has a relatively high concentration of iron (Fe), with detectable elemental signatures for lead (Pb), zinc (Zn), arsenic (As), copper (Cu), and trace amounts of germanium (Ge). It was selected for its high spectral complexity, stemming from both its substantial iron content and the variety of other elements present. The second sample is the NIST reference glass 1412a, which contains elements such as aluminum (Al), calcium (Ca), magnesium (Mg), silicon (Si), strontium (Sr), zinc (Zn), lithium (Li), and boron (B), along with trace elements including barium (Ba), antimony (Sb), lead (Pb), cadmium (Cd), and silver (Ag). The full list of elements and their concentrations can be found in supplementary table 1. This sample was selected for its diverse range of elements at varying concentration levels, allowing us to assess the ability to detect all elements present and to verify the absence of false positives. Next, we demonstrate a direct application to imaging data analysis using a geological sample rich in tin (Sn). This cassiterite ( $\text{SnO}_2$ ) specimen was collected from the margin of a mineralized vein in the Piaotang mine, China. [33].

## 3. ALIAS methodology

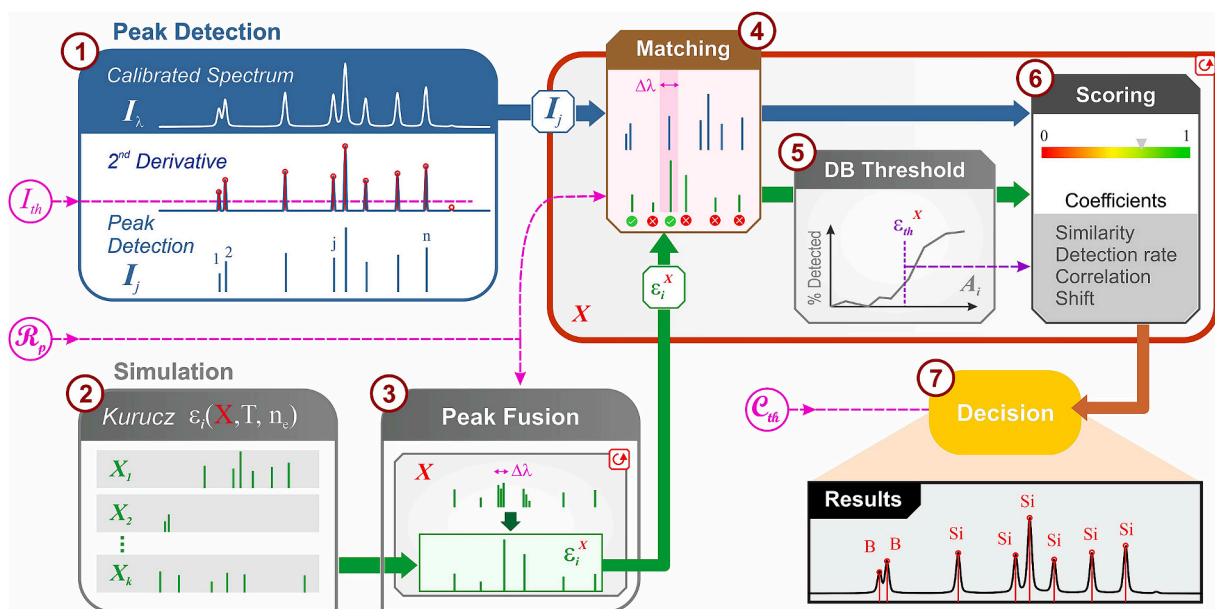
### 3.1. Global overview

The ALIAS methodology was developed with the following objectives: i) to attribute a maximum number of lines to targeted elements with the highest possible reliability, regardless of the level of elemental concentration (major, minor, or trace); ii) to detect potential sources of interference to optimize the choice of the line used during univariate extraction; iii) to minimize computation time to ensure a rapid response during the processing step (ideally less than one second); vi) to limit the number of user parameters as much as possible to simplify its use and make the approach all the more robust, and v) to provide the user with a

confidence level regarding the presence of each detected element. In the version presented in this article, ALIAS operates on low-noise spectra (averaged over a large number of spectra) and with spectra perfectly calibrated in wavelengths.

The general ALIAS methodology is illustrated in Fig. 1. It comprises a total of seven steps executed sequentially, which will be detailed in the following sections. This number of steps may initially seem substantial, but it ensures the robustness of the approach. We subsequently provide the main objectives of these steps along with the definitions of the various terms used:

- 1) Peak Detection:** From the instrumental spectrum  $I_\lambda = f(\lambda)$ , a peak detection is performed using the second derivative method. This operation generates a peak spectrum  $I_j$  containing a total of  $n$  peaks. The parameter  $I_{th}$  allows the user to set the intensity threshold above which the lines will be considered in the peak detection process. In general, this parameter is set to 10 times the baseline noise of the spectrum assumed to be known.
- 2) Theoretical Emissivity:** The theoretical line emissivities are calculated from atomic parameters retrieved from a database (NIST or Kurucz) using a simplified plasma model (plasma in local thermodynamic equilibrium (LTE), without considering Stark effects nor self-absorption). These simulations are carried out for all the elements  $X$  present in the database and for one or more pairs of plasma temperature  $T$  and electron number density  $n_e$ , within temperature and electron number density ranges representative of a laser-induced plasma (typically  $8000\text{K} < T < 12000\text{K}$ , and  $3 \times 10^{16}\text{cm}^{-3} < n_e < 3 \times 10^{17}\text{cm}^{-3}$ ). They are performed only within the spectral range covered by the experimental spectrum. This step allows for the generation of discrete emissivity spectra  $\epsilon_i^X$  of relative amplitude for all the elements  $X$ ,  $i$  being the index of the line.
- 3) Peak fusion:** This step involves grouping the spectral lines from the database based on the resolution power  $\mathcal{R}_p$  of the spectrometer being used. This simplifies the structure of the emissivity spectra  $\epsilon_i^X$  by merging nearby lines together, making it easier to process data.
- 4) Line Matching:** The algorithm checks whether the lines from the theoretical discrete spectra  $A_i^X$  correspond to an experimental peak. This correspondence is established by verifying that the wavelengths align, considering the resolution power  $\mathcal{R}_p$  of the spectrometer.
- 5) DB Threshold:** A graph showing the percentage of detected lines versus the threshold coefficient  $\mathcal{C}_{th}$ . The curve shows a sharp increase in detection rate as the threshold coefficient increases.
- 6) Scoring:** A color scale from 0 (blue) to 1 (red) representing the confidence level. The scoring is based on the following coefficients: Similarity, Detection rate, Correlation, and Shift.
- 7) Decision:** A final step where the results are presented, showing the detected elements (B, Si, Sn) and their corresponding peaks in the spectrum.



**Fig. 1.** Global methodology. The steps 1 to 7 are discussed in the text. Note that parameters  $\mathcal{R}_p$  represents the resolution power of the spectrometer,  $I_{th}$  the intensity threshold applied to the detection algorithm, and  $\mathcal{C}_{th}$  the threshold coefficient applied for the identification decision. DB: database.

Consequently, the algorithm ensures that the theoretical lines are sufficiently close to the observed peaks in the experimental data to be considered a match.

- 5) **Database threshold:** An automated thresholding method is then applied to the theoretical emissivities with the aim of removing all lines from the database that have an emissivity too low to be detected. It also determines the threshold value  $\varepsilon_{th}^x$  of detectable emissivities for each of the elements. This step is crucial for eliminating unnecessary lines and enhancing the performance of the subsequent steps.
- 6) **Scoring:** The purpose of this step is to evaluate several coefficients (such as wavelength similarity, correlation, detection rate, and wavelength shift) which will assess the similarity between the theoretical emissivities and the experimental peak spectrum. All of these coefficients are normalized to yield a value between 0 and 1. The closer the values of these coefficients are to 1, the greater the confidence in the identification of the elements.
- 7) **Decision:** In this final step, we compute for each element a unique coefficient  $k_{det}$  and compare its value to a threshold provided by the user  $\mathcal{C}_{th}$ . If these coefficients exceed the threshold, we consider the element to be present and label the corresponding experimental peaks.

Naturally, this sequence of steps will be applied only within the spectral range of the provided experimental spectrum. To clarify the following discussion, it is useful to define four categories of elements based on the number of lines present within this spectral range: No-Line Element (NLE): no lines present; One-Line Element (OLE): only one line present; Sparse-Line Element (SLE): a few lines present (between 2 and 12); Multiple-Line Element (MLE): a large number of lines ( $> 12$ ). In the following sections, we will delve deeper into the mechanics of each step of the method.

### 3.2. Peak detection

The aim here is to extract the wavelength and intensity of peaks observed in the experimental spectrum using the second derivative method. First, the second derivative of the experimental spectrum is calculated. Next, the negative of the second derivative is taken, with all negative points removed (i.e., set to zero), resulting in a spectrum that enhances the visibility of peaks. A final peak detection is then performed on this modified spectrum, allowing for precise identification of wavelengths and intensities. In the LabVIEW version, peak detection is conducted using the 'Peak Detector' function, whereas in Python, the 'scipy.

signal.find\_peaks' function is employed in prominence mode. This operation generates a peak spectrum  $I_j$ , as a form of a vector, consisting of a total of  $n$  peaks, each represented by a pair of intensity and wavelength position values. The threshold value  $I_{th}$ , which may be adjusted by the user, is applied for retaining only the meaningful information. This method has the advantage of filtering out slow variations in the signal, such as the continuum emission, while providing high precision in peak localization, even for weak lines and for resolving overlapping lines. However, it is very sensitive to the spectrum noise [27]. Nonetheless, this method is well-suited to our need for processing averaged spectra, as after a certain number of averages, noise becomes virtually undetectable in the spectrum. For example, in the instrumental configurations used for  $\mu$ LIBS, the noise measured on the baseline of a typical single-shot spectrum is around 20 counts. After averaging 10,000 spectra, this noise becomes nearly imperceptible, decreasing to a value that is approximately 100 times lower.

Fig. 2 illustrates this process using the example of chalcopryite-tennantite rock, showing the experimental spectrum in black, the negative of the second derivative (with negative values set to zero) in blue, and all detected peaks highlighted in red, totaling 85 peaks. The experimental spectrum represents the average of all the dataset, containing 500,000 single-shot spectra. As can be seen, every visible peak is well detected. Moreover, this method is highly effective in identifying both peaks with significant overlap and small lines near the continuum, as illustrated in the zoomed-in section of the spectrum.

### 3.3. Theoretical emissivity

The calculation of theoretical line emissivities is based on an optically thin plasma in LTE, characterized by a temperature  $T$  and an electron number density  $n_e$ . Additionally, wavelength shifts induced by the Stark effect are not considered. It is important to emphasize that, to reduce computation time, we do not aim to simulate synthetic spectra with a detailed physical or experimental profile of the lines. Instead, the aim is to generate a theoretical discrete spectrum from atomic databases, based on a simple plasma model ( $T$  and  $n_e$ ) with parameters easily adjustable by the user. Under these hypotheses and by using the Saha and Boltzmann equations, the relative emissivity of a line between a high ( $i$ ) and a low ( $j$ ) energy level can be expressed as [34]:

$$\varepsilon_{ij}^q(\lambda, T, n_e) = W^q(T, n_e) \bullet A_{ij} g_i \bullet \frac{e^{-E_i/kT}}{U_s^q(T)}, \quad (1)$$

where  $q$  ( $= I$  or  $II$ ) is the ionization state,  $W^q$  is the probability of

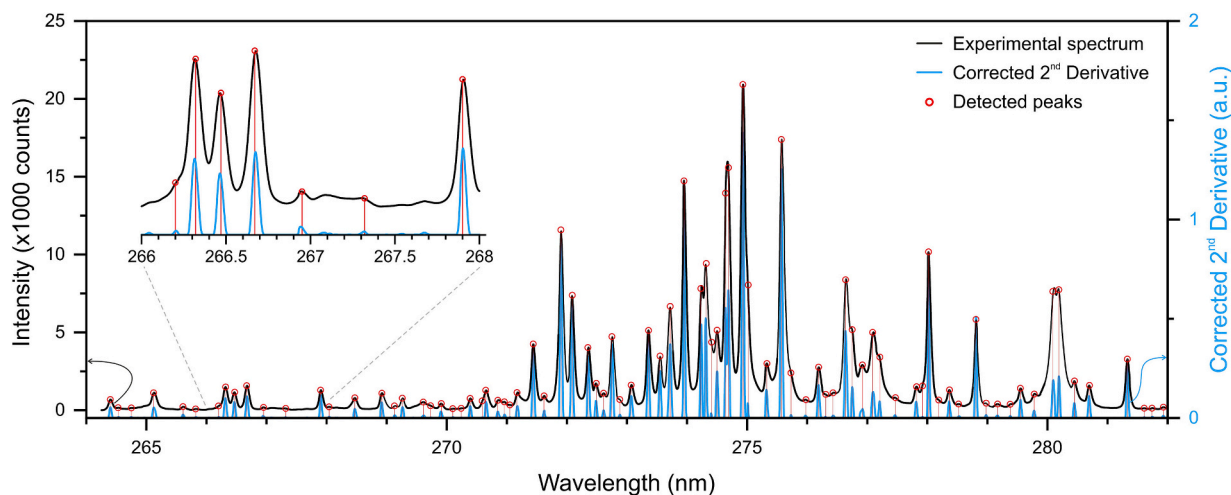


Fig. 2. Peak detection methodology. The proposed experimental averaged spectrum was acquired from the chalcopryite-tennantite sample with a diffraction grating of 2400 lines/mm (spectral resolution: 0.04 nm), containing a high iron content, with the presence of zinc, copper, arsenic, lead and germanium.



occurrence of the ionization state  $q$ ,  $i$  and  $j$  are respectively the index of the higher and lower energy levels,  $A_{ij}$  the Einstein coefficient,  $g_i$  is the statistical weight,  $E_i$  is the energy of the higher level,  $k$  the Boltzmann constant, and  $U_s^q$  is the partition function. The probability  $W^q$  is obtained by considering the Saha equation which gives the ratio between the ion and atom number density:

$$W^I = \frac{\xi}{1 + \xi} \text{ and } W^{II} = \frac{1}{1 + \xi}, \text{ with } \xi = \frac{n^{II}}{n^I} \text{ and } n^{II} = n_e \quad (2)$$

In this article, all emissivity calculations were conducted using atomic parameters provided by the Kurucz database [35]. However, this procedure is also compatible with other databases, such as the NIST database. The Kurucz database was chosen because it provides a list of comprehensive atomic parameters for a large number of elements (79 in total) without missing significant lines, as is sometimes the case for certain elements in the NIST database. The full list of elements and species provided by this database is shown in supplementary table 2. In addition, only neutral atoms and singly ionized ions are considered. These simulations conducted in the experimental spectral range result in a discrete emissivity spectrum  $\epsilon_i^X$  as a form of a vector for each element  $X$ . It consisting of a total of  $N^X$  lines, each represented by a pair of emissivity and wavelength values. To reduce computation time, only one set of temperature and electron number density was used ( $T = 9000 \text{ K}$  and  $n_e = 1 \times 10^{17} \text{ cm}^{-3}$ ) although it is entirely possible to define multiple combinations of these parameters.

### 3.4. Peak fusion

For MLE category elements (those with many lines), it's common to encounter a dense cluster of lines, separated by distances much smaller than the spectrometer's spectral resolution or the Stark broadening of the lines. Because we are working with discrete spectra, this can sometimes lead to notable discrepancies between theoretical data and experimental peaks. A classic example is illustrated in supplementary fig. 1 for iron in the 300 nm spectral range. In this range we can notice two intense experimental lines near 302 and 310 nm (highlighted in green) caused by the superposition of numerous lower-intensity lines. The 'Peak Fusion' step addresses this by summing the intensities of closely spaced lines to create a database peak structure that closely matches the experimental spectrum. The process starts with the more intense line in the spectral range, creating a bin that extends from its position to that position plus and minus the spectrometer's resolution,  $\pm \Delta\lambda/2 = \pm \lambda/(2\mathfrak{R}_p)$ , with  $\mathfrak{R}_p = \lambda/\Delta\lambda$ . All intensities within this bin are summed into a new line, with the position set to that of the strongest emission line within the bin. This procedure is then repeated from the end of the current bin to initiate the next, continuing until the entire spectral range is covered.

### 3.5. Line matching

This step involves checking which theoretical lines correspond to the lines in the instrumental spectrum as proposed in [25]. For each element  $X$  in the database and for each theoretical line, we verify iteratively if their position is within  $\pm \Delta\lambda/2$  of the position of an instrumental peak. If this condition is met, the line is marked as detected, and the position and intensity of the instrumental peak are associated within the emissivity data  $\epsilon^X$ .

### 3.6. Database threshold

In the implementation of ALIAS, we assume that the theoretical calculation of emissivities is similar, or at least very close, to the line emission observed on the experimental spectrum. Indeed, the higher the emissivity of a line, the greater its probability of detection. Therefore, it is essential, from our perspective, to establish an emissivity threshold

below which the lines of an element cannot be detected. This step is indeed crucial for eliminating lines that are unlikely to be detectable due to low emissivity, which would prevent them from appearing in the experimental spectrum. This is particularly important for the MLE category, as these elements can exhibit a substantial number of undetected lines that may significantly skew the calculation of the coefficient values (see the next sections). It is essential to emphasize that the goal is not to establish an exact threshold, but rather to remove as many lines as possible while ensuring that no detectable line is inadvertently discarded.

To illustrate, we will detail the proposed 'database threshold' using two MLEs: one detected in the instrumental spectrum (Fe) and the other not detected (Mn), in the case of the chalcopyrite-tennantite sample. Fig. 3a and b show the high density of theoretical lines in this spectral region (264–282 nm) for Fe and Mn, respectively. In these figures, the instrumental spectrum displayed in logarithmic scale has been added as a dotted gray line for illustration purposes (spectrum from Fig. 2). In each figure, an inverted triangle indicates each detected line (i.e., those satisfying the line-matching criteria outlined in Section 3.5). For iron, 65 lines are detected out of the 259 presents in this spectral range, while for manganese, 40 out of 240 lines are detected. For each element, all theoretical emissivity values are sorted in ascending order, and a graph is created to represent each detected or non-detected line based on its emissivity value. These results are illustrated in Fig. 3c and d for Fe and Mn, respectively. As can be seen, the number of detected lines for iron appears to be significantly higher as the emissivity of the lines increases, whereas a different behavior is observed for manganese. In the case of Mn, detected lines are indeed very scattered across the entire range of emissivity values. This underscores the fact that despite 40 detected lines for Mn, this element is likely not present in the instrumental spectrum.

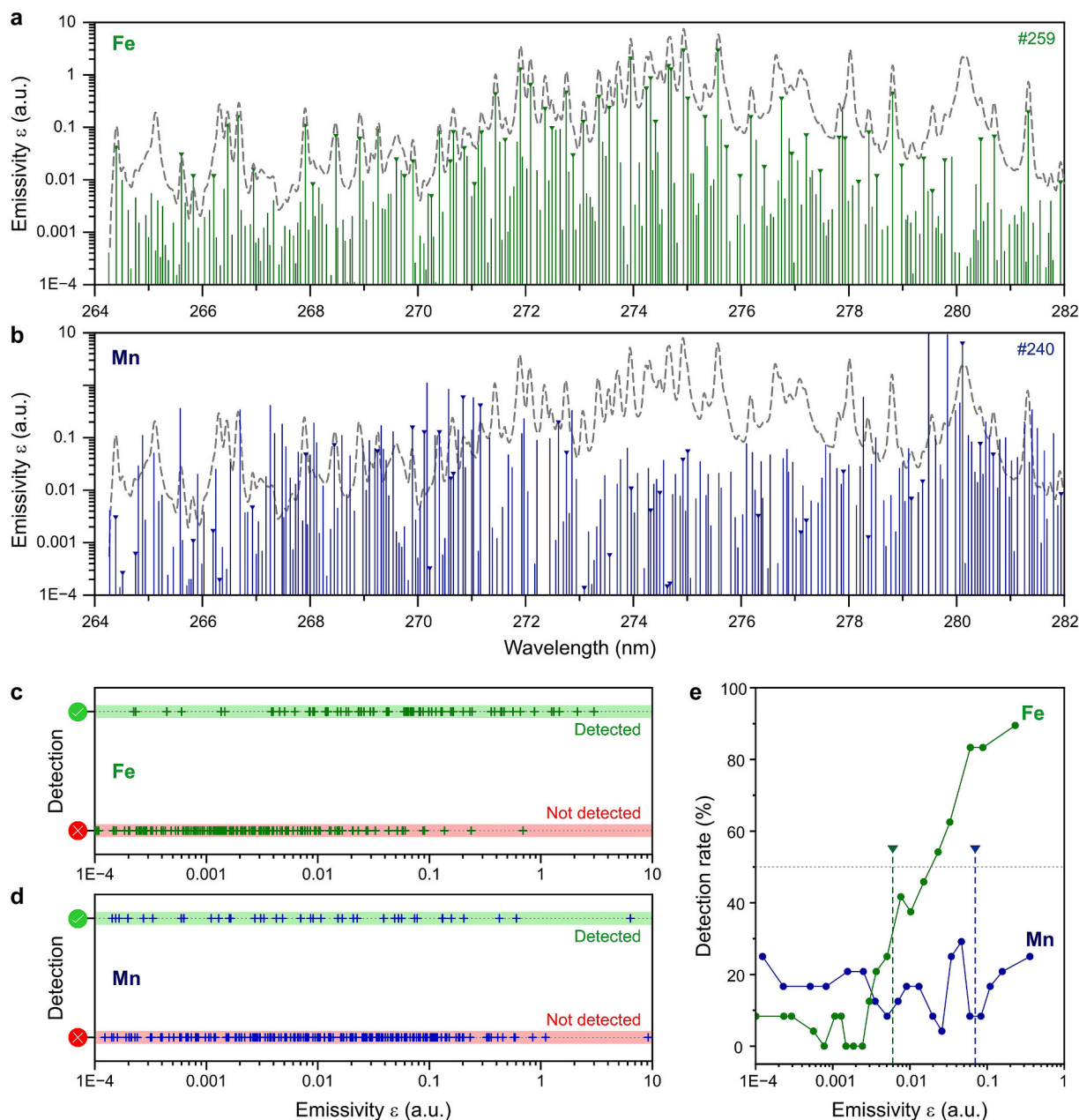
To qualify this, the emissivity scale is divided into several bins, each typically corresponding to a width of 10 lines for MLE (and 4 lines for SLE). For each of these bins, we evaluate the percentage of detected lines (defined simply as the number of detected lines divided by the number of lines in the bin). As shown in Fig. 3e, such detection rates are then plotted as a function of the averaged emissivity value for each bin. If the rate exceeds a value of 50 %, as is the case with iron, we define the threshold as the minimum emissivity value of the segment that yields a detection rate of 30 %. Conversely, for manganese, we set the threshold to retain only the most intense lines, based on a decade from the highest emissivity value. Any lines with emissivity values below the threshold established for each element will not be considered in the following. In the example illustrated in Fig. 3, only 138 out of 259 lines are retained for iron, and 54 out of 240 for manganese.

### 3.7. Scoring

In this step, the objective is to calculate various coefficients that will enable a decision on the presence or absence of each element. Several coefficients are necessary to best adapt to the situations encountered, particularly different categories of elements (MLE, SLE, and OLE). All these coefficients are defined to yield a value between 0 and 1. The closer the values of these coefficients are to 1, the greater the confidence in the identification of the elements.

#### a) Cosine Similarity

The coefficient 'cosine similarity', is as instance widely used in mass spectrometry for identification of experimental spectra against libraries of spectra of known compounds [36]. The idea is simply to convert both discrete peak spectra for each element (theoretical  $\epsilon_i^X$  and experimental  $I_i$ ) into vectors and calculate the cosine of the angle between these two vectors [29]:



**Fig. 3.** Database threshold principle illustrated for Fe and Mn in the case of the chalcopyrite-tennantite sample in the range 264–282 nm. a) Database lines for Fe. b) Database lines for Mn. The experimental averaged spectrum is displayed as dot lines. c) Fe lines detected. d) Mn lines detected. e) Detection rate v.s. database emissivity.

$$k_{sim}(X) = \frac{\sum_{i=1}^n \epsilon_i^X \times I_i}{\sqrt{\sum_{i=1}^n \epsilon_i^{X^2}} \sqrt{\sum_{i=1}^n I_i^2}} \quad (3)$$

where  $n$  is the total number of peaks detected in the case of the element  $X$ . This coefficient can be considered a score indicating how similar two vectors (or spectra) are. Cosine similarity is a powerful tool for comparing spectra by focusing on shape matches, regardless of absolute intensity variations. The cosine of the angle between two spectral vectors provides a similarity measure that is independent of vector length, enabling the identification of spectra with similar features even when their intensities differ. This coefficient will be all the more robust as the number of lines increases and as the emissivity simulations are accurate. Some additional information about this coefficient calculation is provided in the supplementary materials section D.

#### b) Detection Rate

The main idea here is to determine, for each element, the ratio between the number of detected lines and the total number of lines theoretically detectable in the spectrum. To give more weight to high-intensity lines, the  $k_{rate}$  coefficient is defined as a weighted sum based on the theoretical emissivity values of the ‘detected line’ or ‘non-detected line’ ( $t$ ) events:

$$k_{rate}(X) = \frac{\sum_{i=1}^n t_i \times \epsilon_i^X}{\sum_{i=1}^n \epsilon_i^X} \quad (4)$$

with  $t_i$  taking the value 1 if line  $i$  is detected (i.e., corresponds to a peak in the spectrum) and 0 otherwise. This coefficient is particularly useful for elements with a large number of lines (MLE), as it provides a statistical insight into detection. It is also valuable for elements with few

lines' element (SLE and OLE), as a deviation from a value of 1 can indicate whether higher-emissivity lines have not been detected.

### c) Wavelength Shift

The issue with the two preceding coefficients, in particular the similarity coefficient, is that they are not well-adapted for elements with few lines (SLE), or even just a single line (OLE). Given the limited spectral ranges we are working with, such situation is not uncommon. To address this, we propose a third coefficient that measures the experimental wavelength deviation from the theoretical line position. The shift coefficient for an element with a single line is thus defined as:

$$k_{\text{shift}} = 1 - \frac{|\lambda_{\text{th}} - \lambda_{\text{exp}}|}{\Delta\lambda} \quad (5)$$

where  $\lambda_{\text{exp}}$  and  $\lambda_{\text{th}}$  are the positions of the experimental peak and the database line, respectively. In the case of elements with multiple lines, this coefficient is calculated as the weighted averaged of the  $k_{\text{shift}}$  value obtained for each line, with weighting based on emissivity to give greater importance to the more intense lines.

### 3.8. Decision and probability evaluation

As discussed above, the  $k_{\text{sim}}$  coefficient is efficient for element that possesses a significant number of lines (MLE), while the  $k_{\text{shift}}$  coefficient is better adapted for elements with a low number of lines (SLE and OLE). Besides, since  $k_{\text{rate}}$  measures the overall percentage of detected lines, it is useful across all categories of elements. Finally, we combined these three coefficients to define a unique 'detection coefficient'  $k_{\text{det}}$  that can adapt to all situations (OLE, SLE, MLE):

$$k_{\text{det}}^X = k_{\text{rate}}^X \times \left( \frac{1}{N_X} k_{\text{shift}}^X + \frac{N_X - 1}{N_X} k_{\text{sim}}^X \right) \quad (6)$$

where  $N_X$  is the total number of detected lines for the element  $X$ . Note that using  $N_X$  in the term within parentheses simply allows for a linear combination of the  $k_{\text{shift}}$  and  $k_{\text{sim}}$  coefficients, assigning greater weight to the similarity and lower weight to the shift coefficient as the number of lines is increasing. The final decision on whether element  $X$  is detected or not simply involves comparing the value of  $k_{\text{det}}$  to the threshold value  $\mathcal{E}_{\text{th}}$  defined by the user. If  $k_{\text{det}} > \mathcal{E}_{\text{th}}$ , then element  $X$  is retained and indicated as present. The threshold value is set by default to 0.5. It is important to mention that these definitions were established empirically by conducting tests in different spectral configurations. In cases where multiple plasma parameter pairs ( $T$ ,  $n_e$ ) are used in simulations, the best pair will be chosen as the one yielding similarity coefficients closest to unity.

Additionally, alongside the  $k_{\text{det}}$  coefficient, we defined a confidence level  $CL$  (or detection probability). A detailed calculation of this confidence level is provided in the supplementary materials section E. Briefly, it is assessed by combining four factors: (i) the  $k_{\text{det}}$  coefficient value, (ii) the detection or absence of a major line in the spectral range, (iii) the signal-to-noise ratio of the peaks, and (iv) the rarity of the element in question based on known values for Earth's crust. This probability coefficient (or confidence level) is not essential for the proper ALIAS functioning. However, in certain edge cases (when  $k_{\text{det}}$  coefficient is low), it provides additional information that may help the user decide whether a given element is present or absent.

### 3.9. Algorithm code

The proposed ALIAS methodology was implemented in two different programming languages: Python and LabVIEW. The Python version utilizes the 'Spyder IDE' and leverages standard libraries such as 'NumPy', 'Pandas', 'SciPy', and 'Matplotlib'. All processing was conducted on laptops equipped with the following specifications: Intel i9 processor,

32 GB of RAM, and 1 To SSD storage.

## 4. Results

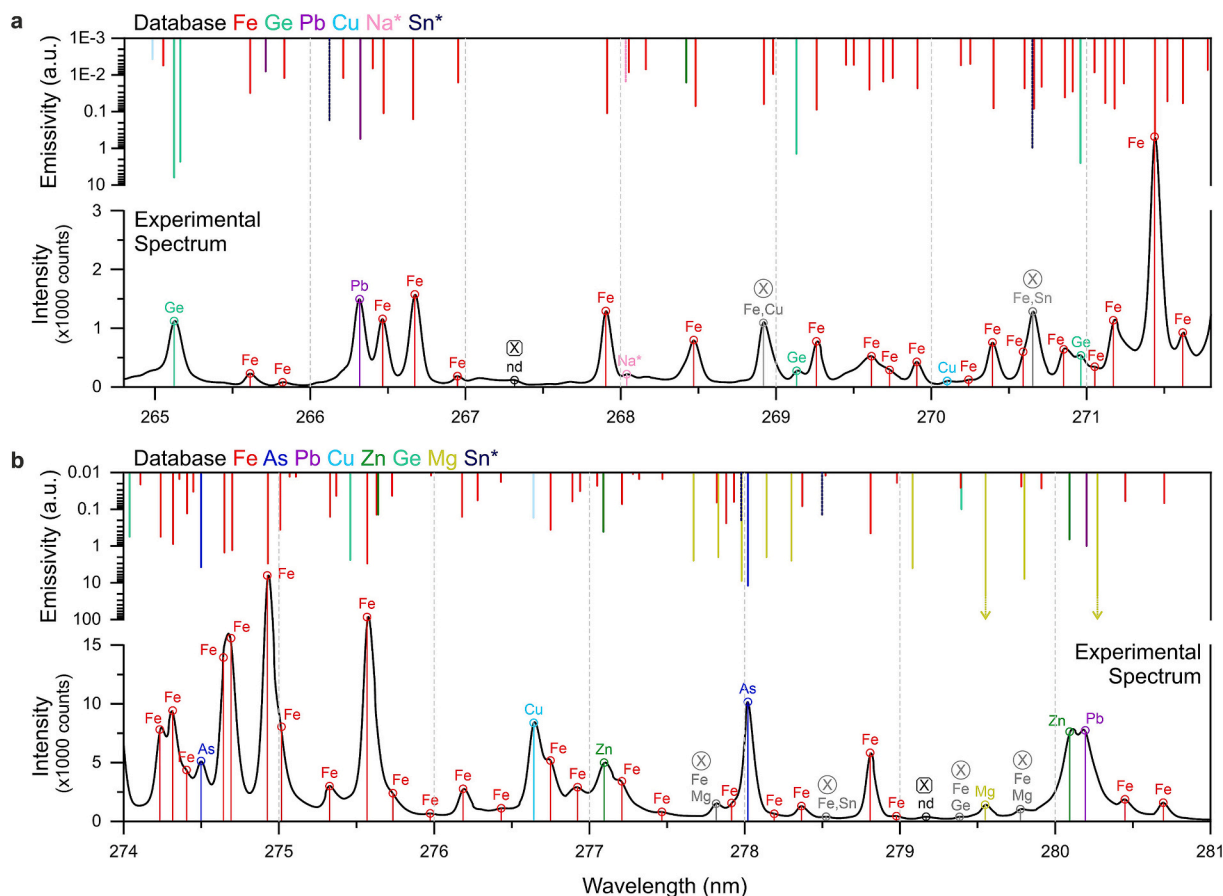
### 4.1. Chalcopyrite-tennantite sample

This chalcopyrite ( $\text{CuFeS}_2$ ) and tennantite (ideal formula  $[\text{Cu}_6[\text{Cu}_4(\text{Fe,Zn})_2]\text{As}_4\text{S}_{13}]$  sample was selected to evaluate ALIAS performance with a highly dense spectral structure caused by a high iron concentration. In addition to chalcopyrite and tennantite, quartz, galena and rutile, are potentially present. Therefore, in the processed average spectrum, we expect to detect Fe, as well as possible lines for As, Pb, Zn, and Cu, among others. Furthermore, geochemical analyses indicate that this sample contains traces of Ge. The spectral range was chosen specifically to facilitate Ge detection, as it includes this element's main line at 265.1 nm. These characteristics make it a very interesting sample, significantly richer than most iron alloys. Another noteworthy aspect is that this spectral range contains a wide range of iron lines with varying emissivities, including a substantial number of weak lines, which may present challenges in term of database accuracy. The identification results obtained by applying ALIAS to the chalcopyrite-tennantite spectrum shown in Fig. 2 are indicated in Fig. 4. The complete set of coefficients obtained are listed in supplementary table 3 and a screenshot of the LabVIEW code version is provided in supplementary fig. 4. Due to the large number of lines in this spectral range (250 to 280 nm), we chose to illustrate the identification results in two regions: 265–272 nm and 274–281 nm, shown in Fig. 4a and b, respectively. The parameters used for ALIAS were  $I_{\text{th}} = 90$  counts,  $\mathcal{E}_{\text{th}} = 0.5$ , and  $\mathfrak{N}_P = 6500$ . The bottom part of each figure represents the experimental spectrum (solid black) as well as all detected lines with assigned element names. The top part represents the emissivities, calculated with the atomic parameters provide by the Kurucz database, of the detected element displayed in a reverse and logarithm scale. A specific color has been assigned for each detected element.

Additionally, potentially interfering lines are shown in gray, and undetected lines are marked in black with the 'nd' symbol. As seen, almost all experimental lines have been identified, with only two lines undetected ('nd') out of the 64 found in Fig. 4a and b, representing a nearly 97 % identification rate. All expected elements are present, consistent with our knowledge of the sample. In addition to Fe, Ge, Pb, Zn, Cu, and As, ALIAS identifies ( $k_{\text{det}} > \mathcal{E}_{\text{th}}$ ) three additional elements, including Mg, Na, and Sn. While Mg seems entirely plausible with, the accuracy of identification for Na and Sn is questionable. It is worth noting that these two elements have a detection coefficient close to 0.5 and are marked with an asterisk (\*) to indicate a confidence level  $CL$  below 50 %. Na has only a single weak line in this spectral range, with low emissivity. The situation for Sn is different, as two lines are detected, but both are interfered with iron (c.f. Fig. 4). We took a closer look at the case of the undetected lines. It appears that these two lines are associated with iron lines whose intensities are significantly underestimated in the Kurucz database.

### 4.2. SRM1412 reference glass

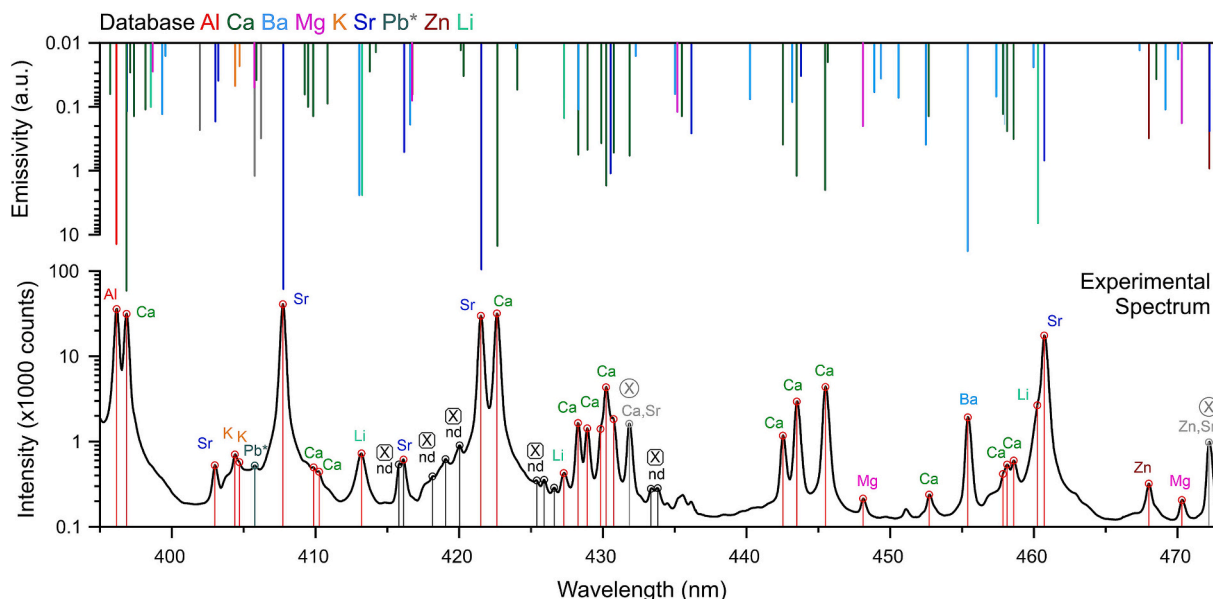
The SRM 1412 reference glass was chosen because it is a multi-component glass typically used for validating chemical analysis methods. It contains various elements at different concentrations, making it well-suited for validating analytical techniques. The list of elements present and their associated concentrations is provided in supplementary table 1. It contains around twenty elements, with concentrations ranging from a few tens of percent to a few tens of ppm. Multiple analyses were performed across different spectral ranges and at various spectral resolutions. For each range, a sequence of  $100 \times 200$  spectra was acquired with a lateral resolution of 20  $\mu\text{m}$ , and the corresponding averaged spectrum was computed. The identification results provided by ALIAS on the averaged spectrum corresponding to the



**Fig. 4.** Identification results for the chalcopyrite-tennantite rock sample shown in Fig. 2. For better clarity due to the large number of lines, two significant spectral regions are shown: a) from 265 to 272 nm and b) from 274 to 281 nm. Note that elements (Sn and Na) with an asterisk (\*) correspond to confidence level lower than 0.5.

396–470 nm spectral range (600 lines/mm grating) are shown in Fig. 5. The other probed spectral ranges and the corresponding ALIAS results are provided in supplementary fig. 5, which displays screenshots from the LabVIEW software. The bottom part in Fig. 5 represents the experimental spectrum (solid black) as well as all detected lines with assigned

element names. Experimental intensities are displayed in a logarithmic scale to better illustrate the measurement dynamics. The top part represents the Kurucz database emissivities of the detected element displayed in a reverse and logarithm scale. A specific color has been assigned for each detected element. The parameters used for ALIAS in



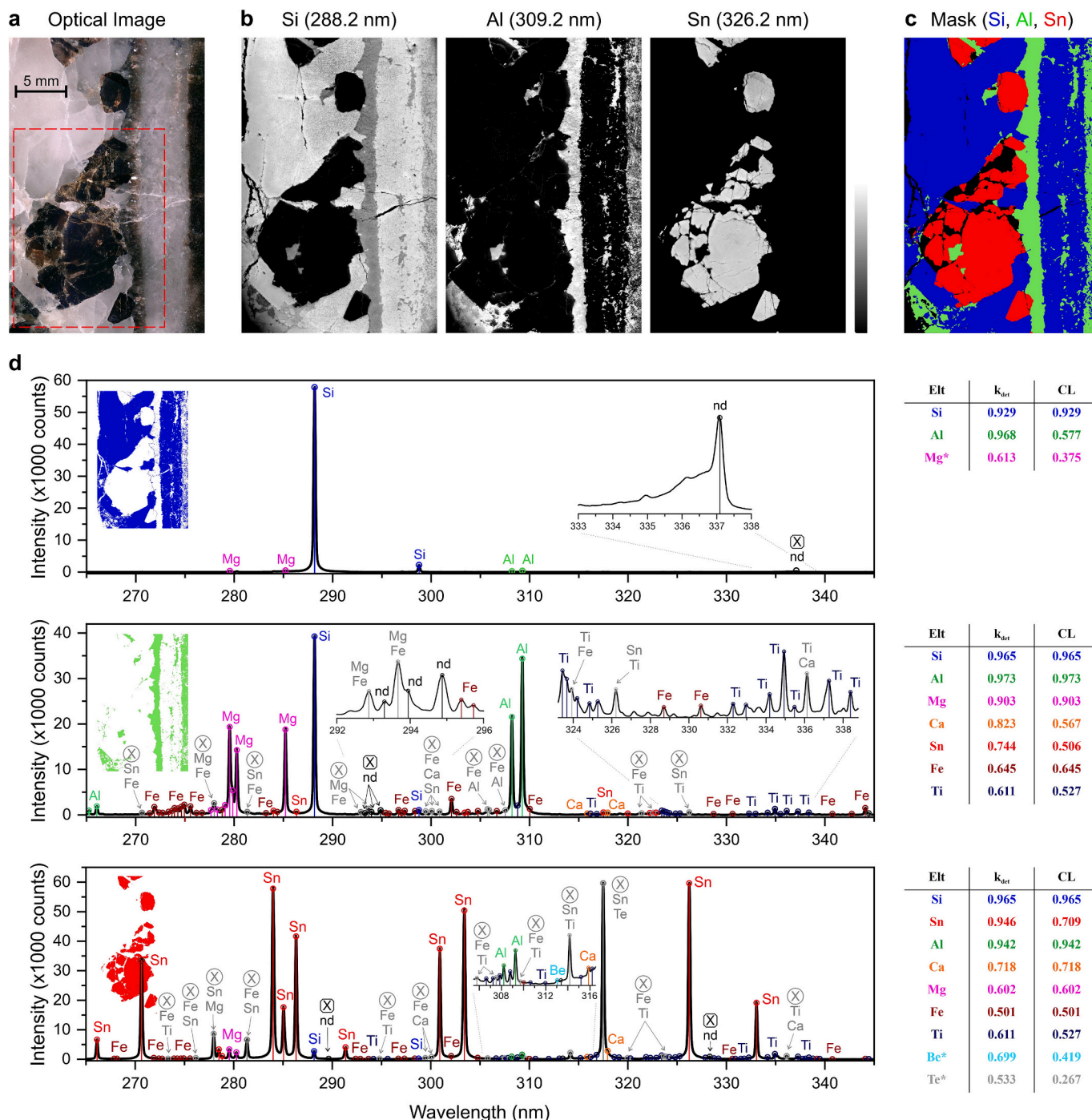
**Fig. 5.** Identification results for the SRM1412 Reference Glass in the spectral region 395–470 nm.



this case were  $I_{th} = 100$  counts,  $\mathcal{E}_{th} = 0.5$ , and  $\mathfrak{N}_P = 2500$ . As previously, potentially interfering lines are shown in gray, and undetected lines are marked in black with the 'nd' symbol. The computation time required for the identification of this spectrum was less than 1 s.

As can be seen in Fig. 5, almost all lines have been identified, with a total of 9 elements detected in this spectral range: Al, Ca, Sr, K, Zn, Mg, Li as major elements, Ba and Pb as minor/trace elements. All expected elements emitting in this spectral range are present, consistent with our knowledge of the sample. A total of only eight low-intensity lines remain

unidentified. Upon verification, these lines do not match any elements in the database and are most likely due to unidentified weak molecular emissions. Overall, ALIAS performs well, with no incorrect identifications. Additionally, it handles the measurement dynamics (spanning approximately three orders of magnitude), identifying both intense lines and very weak ones, such as Pb around 405 nm. Finally, the identification of a Li line at the base of a prominent strontium peak near 460 nm underscores the advantage of employing peak detection based on the second derivative of the spectrum.



**Fig. 6.** Example of ALIAS application to a LIBS imaging dataset. a) Optical image of the analyzed sample surface. b) Elemental maps of the major elements. c) Phase masks generated by thresholding the major elements. d) Averaged spectra corresponding to the three phases identified in (c) and ALIAS results. The non-detected element 'nd' are displayed in black, while the lines suffering from potential interference are indicated in gray. The red dashed rectangle in (a) outlines the area corresponding to the elemental images shown in Fig. 7. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The analyses conducted on this sample enabled the detection of 15 elements present in the glass (Si, Al, Sr, K, Zn, Na, Ca, Mg, Li, B, Ba, Pb, Sb, Ag, and Cd). Only As, Cr and Fe were not detected, likely due to their low concentrations and the selected spectral ranges, which were not necessarily optimal for their detection. More importantly, no mis-identifications were observed, despite the presence of numerous low-intensity molecular emissions polluting the spectral baseline.

#### 4.3. Application to LIBS imaging processing: cassiterite rock

As a final test, we propose applying ALIAS to process a LIBS imaging dataset acquired on a sample from the Piaotang mine (China) [33]. It was collected from the margin of a mineralized vein. The main mineral phase in the sample has been identified as cassiterite ( $\text{SnO}_2$ ), illustrated in Fig. 6a. This dataset contains 2 million spectra recorded using a 600 lines/mm grating, covering a spectral range of approximately 80 nm between 265 and 345 nm with a resolution power  $\mathfrak{R}_p \sim 2500$ . The spatial resolution is 20  $\mu\text{m}$ .

The first step of the processing is briefly discussed here. It involves calculating the averaged spectrum from all the dataset spectra. ALIAS is then applied to this averaged spectrum to identify the major elements using a high detection threshold. Results obtained by ALIAS on this global averaged spectrum are illustrated in supplementary fig. 6. Among the detected elements, the most intense lines are selected, and their intensities are extracted from individual spectra to generate the corresponding elemental images [23]. The results are illustrated in Fig. 6b, showing the three major elements detected: Al, Si, and Sn. Our knowledge of the sample, combined with the observed elemental distribution, allows us to clearly define the different mineral phases of the sample, leading to the generation of masks representing the three main mineral phases shown in Fig. 6c with Al in green (aluminosilicate), Si in blue (quartz), and Sn in red (cassiterite) [24]. Note that black areas in Fig. 6c indicate holes in the sample (spectra without signal). By applying these masks to the dataset, it becomes possible to calculate the averaged spectra corresponding to each mineral phase and use ALIAS to identify the elements present in these spectra. These results are illustrated in Fig. 6d. The parameters used for ALIAS were  $I_{th} = 100$  counts,  $\mathcal{C}_{th} = 0.5$ , and  $\mathfrak{R}_p = 2500$ .

In the phase defined by Si (top spectrum), ALIAS identifies the two Si lines present in this spectral range, as well as low-intensity lines associated with trace levels of Al and Mg. The predominance of Si strongly suggests a ‘quartz’ phase. It is worth noting that the line labeled ‘nd’ around 337 nm appears to be an unidentified molecular emission. For the phase defined by Al (middle spectrum), a significantly higher number of lines is observed. The predominance of Si, Al, and Mg suggests the presence of an aluminosilicate phase. Other elements, associated with much weaker intensity lines, such as Fe, Ca, Ti, and Sn, are also identified with acceptable confidence levels. This example illustrates the potential of ALIAS to assist in the identification of elements in spectrally and dynamically complex spectra. It also shows its usefulness in highlighting lines that may be affected by spectral interferences (indicated in gray in Fig. 6). This feature can help guide the selection of the most appropriate lines for generating elemental images of the target elements. In this example, an unidentified triplet is observed around 294 nm. These three lines, most likely attributed to Mn, highlight a limitation of the method. ALIAS does not detect Mn because, in this spectral range, the most intense Mn lines are located near 280 nm, a region dominated by strong Mg lines. In this case, the predominance of Mg masks Mn’s intense lines, resulting in a detection coefficient  $k_{det} < 0.5$ . However, it is worth noting that Mn appears just below Ti in the list of elements provided by ALIAS, with a detection coefficient  $k_{det} = 0.41$ .

Finally, for the last phase, characterized by a strong dominance of Sn (cassiterite), which is the primary focus of this study, the ALIAS results are shown on the bottom spectrum. As expected, a predominance of Sn is observed, along with several elements displaying low-intensity lines,

such as Fe, Al, Ti, Si, Ca, and Mg. Additionally, Be and Te are detected with confidence levels below 0.5, as both elements only have one spectral line within the studied range. For Be, the observed line around 313 nm, which is free from interference with other elements, is worth considering. In contrast, for Te, whose line is heavily interfered with by those of Sn, the dominant element in this phase, the interpretation is far more uncertain. This highlights a key limitation: when an element has only one line that perfectly matches the wavelength of another element’s line, ALIAS will report it as detected. This is one of the reasons why we introduced a confidence level that incorporates multiple factors (see supplementary material section E) in addition to the  $k_{det}$  coefficient. Based on ALIAS results, non-interfered lines were selected for each element, enabling the extraction of intensities and the generation of elemental maps corresponding to the cassiterite phase. These results, illustrated in Fig. 7, reveal two distinct growth line zonations for Ti and Fe, along with a highly localized Be concentration in a restricted zone.

## 5. Discussion

The ability of ALIAS to identify spectral lines within narrow spectral ranges and across a wide intensity dynamic range has been illustrated through several representative examples. The results are encouraging, showing good identification rates and, importantly, a high level of reliability. In addition, the method provides useful indications of potential interferences with other elements, which can help in selecting the most appropriate lines for elemental image extraction. These features suggest that ALIAS could become a helpful tool for LIBS spectrum analysis, potentially saving time and reducing line attribution errors. Although the current version of the method already meets the expected objectives, we will further discuss possible improvements and address several aspects that have so far received limited attention.

It is first of all essential to emphasize that ALIAS operates with spectra that are perfectly calibrated in wavelength. This calibration step can be problematic in certain cases and requires the prior use of a calibration source (e.g., Hg—Ar) or the analysis of a known reference sample. In this work, we typically used brass or tungsten sample to calibrate the spectrometer before data acquisition. Similarly, it is important to note that in the proposed version, ALIAS is not adapted for cases where the Stark shift is too significant, particularly for spectra recorded at very short delays or for lines with exceptionally high shift coefficients, since the  $k_{shift}$  coefficient will inevitably be degraded. In this regard, a possible improvement would be to incorporate Stark shift and broadening coefficients into the database parameters. From a technical standpoint, accounting for the influence of the Stark effect in emissivity calculations is relatively straightforward. However, the main practical limitation is the limited availability and documentation of Stark broadening coefficients in the literature.

Besides, the computation times are primarily influenced by the number of lines from the database that need to be processed within the spectral range of interest, typically 0.5 s for a spectrum spanning 20 nm and 1 s for one spanning 80 nm. This timescale offers a near-instantaneous response from a user’s perspective, but remains far too long to support the processing of single-shot spectra in the context of LIBS imaging. One possible way to reduce computation time is to restrict the number of elements and/or spectral lines considered in the database. However, such a simplification would likely come at the cost of reduced identification performance and reliability.

It should also be noted that we did not discuss the influence of plasma parameters ( $T$  and  $n_e$ ) on the ALIAS performance. However, it is worth mentioning that in all tests conducted at varying these parameters, no significant differences were observed, as already pointed out in the literature [29].

Finally, apart from the plasma parameters used for emissivity calculations, only three parameters remain adjustable by the user: i) the parameter  $\mathfrak{R}_p$  corresponding to the spectrometer’s resolution power. This parameter directly depends on the spectrometer’s configuration

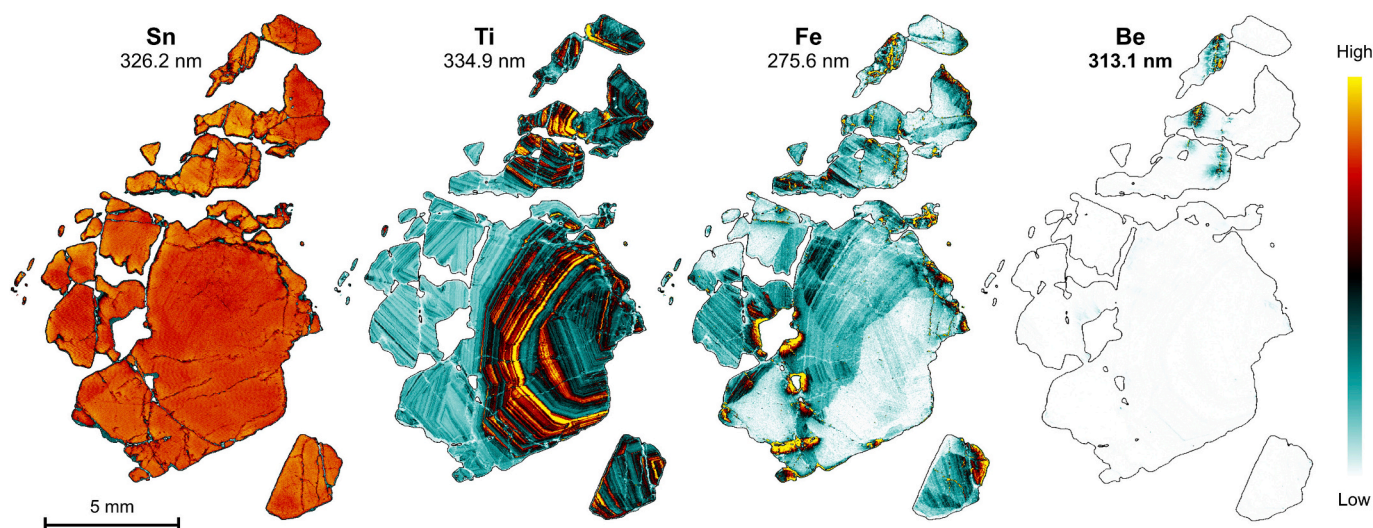


Fig. 7. Elemental images of Sn, Ti, Fe and Be corresponding to the cassiterite mineral phase. Region of indicated in Fig. 6a by the red dashed rectangle. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

which is in our case primarily determined by the diffraction grating selected for the measurement. ii) The decision threshold  $\mathcal{E}_{th}$ , which was set to 0.5 for all the results presented in this article. iii) The intensity threshold  $I_{th}$ , which was ultimately the only parameter we adjusted for the different spectra studied. This detection threshold depends on the spectrum signal-to-noise ratio and could also be fixed if the spectrum noise level is known. The fact that this approach relies on a limited number of parameters, which can be fixed for a given experiment, suggests that ALIAS could eventually establish itself as a universal method for the automatic detection of LIBS spectra, and more generally, atomic emission spectra. Future research should be conducted in this direction to assess its performance across a wide range of experimental configurations.

## 6. Conclusion

In this work, we have presented a methodology for the automatic identification of emission lines in LIBS spectra, referred to as ALIAS. Its capabilities have been illustrated across a range of representative LIBS applications, including reference materials with known compositions, mineral samples, and megapixel-scale imaging. The strength of the proposed methodology lies in its comprehensive, multi-step approach, which combines existing methods from the literature with new strategies aimed at increasing confidence in the identification and validation of detected elements. ALIAS has demonstrated good accuracy and robustness, even under challenging conditions such as wide spectral intensity dynamics or the presence of elements with a large number of lines. Moreover, ALIAS can detect potential spectral interferences, which helps reduce the risk of selecting inappropriate lines for univariate analysis. The code executes within a reasonable timeframe ( $\sim 1$  s), making it suitable for near real-time applications. It is implemented in two widely used programming environments—LabVIEW and Python—which facilitates its integration into diverse experimental setups. Thanks to its solid performance on high dynamic range spectra and narrow spectral ranges, ALIAS appears particularly promising for LIBS imaging, as illustrated through its application to a complex, cassiterite-rich tri-phase rock sample.

Possible future improvements include enriching the database (e.g., by adding molecular emissions, Stark parameters, and/or combining experimental and theoretical data) as well as implementing an automatic calibration step for experimental spectra. Overall, this work contributes to the advancement of automated LIBS spectral processing, with the potential to reduce analysis time, improve identification in

ambiguous cases, and promote standardized data treatment for better comparability across the LIBS community.

## Code availability statement

The ALIAS methodology has no commercial intent. It may be freely re-implemented and improved. Although the code is not publicly hosted, it can be provided for academic purposes upon request.

## CRediT authorship contribution statement

**Clément Noël:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Lana Neoričić:** Writing – review & editing, Supervision, Methodology. **César Alvarez-Llamas:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Alexandre Cugerone:** Writing – review & editing, Validation, Resources, Investigation. **Cécile Fabre:** Writing – review & editing, Validation, Resources, Investigation. **Ludovic Duponchel:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Vincent Motto-Ros:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Funding acquisition, Conceptualization.

## Declaration of competing interest

None.

## Acknowledgement

We kindly thank Hélène Legros for the Cassiterite sample. This work was partially supported by the French region Rhônes Alpes Auvergne (Optolyse, CPER2016), the French “Agence Nationale de la Recherche” (ANR-22-CE27-0017 “MEMOar”, ANR-20-CE17-0021 “dIAg-EM”, ANR-20-CE05-0015 “Micro-Q-Li”, ANR-23-CE42-0023 “AISLE”), the French government “France 2030” initiative, under the DIADEM program managed by the “Agence Nationale de la Recherche” (ANR-22-PEXD-0014, “Libelul”), and the project imATLAS, supported by the European Union’s Horizon Europe Research and Innovation Program under the MSCA Postdoctoral Fellowships 2022, with agreement number 101106090.



## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.sab.2025.107255>.

## Data availability

Data will be made available on request.

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